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পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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SUB-SOIL INVESTIGATION REPORT

CONSTRUCTION OF BASEMENT WITH 14 STORIED RESIDENTIAL CUM COMMERCIAL BUILDING AT GONGANAGOR R/A SOUTH SURMA, SYLHET
DAG-681,682, KHA- 913,914, J.L-107, MOUJA- HOBINONDHI

CLIENT

MR. MASHUK MIAH

Verification Code: D34zA87A
Verify report at: crtc-cee-sust.com

CONDUCTED BY

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

SAMPLE COLLECTED BY

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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would necessary to select proper type of foundation.

Objectives

The purpose of the sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

Scope of work

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

Table – 1: Items of the supplied samples

No	Name	Borehole no	Depth	Quantities
01	Disturbed sample	01	0-70 ft	14
02	Disturbed sample	02	0-70 ft	14
03	Disturbed sample	03	0-70 ft	14
04	Disturbed sample	04	0-70 ft	14
05	Disturbed sample	05	0-70 ft	14
06	Disturbed sample	06	0-60 ft	12
07	Disturbed sample	07	0-70 ft	14
08	Disturbed sample	08	0-70 ft	14
09	Disturbed sample	09	0-70 ft	14
10	Disturbed sample	10	0-70 ft	14
11	Disturbed sample	11	0-70 ft	14
12	Disturbed sample	12	0-70 ft	14
13	Disturbed sample	13	0-70 ft	14
14	Disturbed sample	14	0-70 ft	14
15	Disturbed sample	15	0-70 ft	14
16	Disturbed sample	16	0-70 ft	14
17	Disturbed sample	17	0-70 ft	14
18	Disturbed sample	18	0-70 ft	14
19	Disturbed sample	19	0-70 ft	14
20	Disturbed sample	20	0-70 ft	14
21	Disturbed sample	21	0-70 ft	14
22	Disturbed sample	22	0-70 ft	14
23	Disturbed sample	23	0-70 ft	14
24	Disturbed sample	24	0-70 ft	14

Laboratory Classifications

The disturbed soil samples, supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with borehole (BH) and depth (Please see enclosed Tables).

Laboratory Works & Analysis

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

Bearing capacity for shallow foundation

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)

N-values	Consistency	Unconfined Compressive Strength (q_u)	
		tsf	kPa
0-2	Very soft	0 to 0.25	0 to 25
3-5	Soft	0.25 to 0.50	25 to 50
6-9	Medium	0.50 to 1	50 to 100
10-16	Stiff	1 to 2	100 to 200
17-30	Very stiff	2 to 4	200 to 400
>30	Hard	>4	>400

Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
1 – 2	Very Loose	0 – 15	26 – 28	70 – 100
3 – 6	Loose	15 – 35	28 – 30	90 – 115
7 – 15	Medium	35 – 65	30 – 34	110 – 130
16 – 30	Dense	65 – 85	33 – 38	110 – 140
>30	Very Dense	>85	>50	130 – 150

Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
2 – 3	Very Loose	0 – 15	27 – 28	70 – 100
4 – 7	Loose	15 – 35	30 – 32	90 – 115
8 – 20	Medium	35 – 65	32 – 36	110 – 130
21 – 40	Dense	65 – 85	36 – 42	110 – 140

Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)

N values	Compactness	Relative Density, Dr %	ϕ (degree)	Unit Weight (pcf)
3 – 6	Very Loose	0 – 15	28 – 30	70 – 100
5 – 9	Loose	15 – 35	30 – 34	90 – 115
10 – 25	Medium	35 – 65	33 – 40	110 – 130
26 – 40	Dense	65 – 85	40 – 50	110 – 140
>40	Very Dense	>85	>50	130 – 150

Bearing capacity for deep foundation

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

Table 6: Formula Used to Determine Bearing Capacity of Soil

Shallow Foundation	
For Cohesive Soil	
$q_a = q_u + \gamma \frac{D}{3}$ $q_a = q_u$(Common Practice)	Where, q_a = Allowable bearing capacity, q_u = Unconfined Compressive Strength, γ = Unit Weight of soil, D = Depth of footing
For Cohesion less Soil	
$q_{ult} = 30N$; $q_a = q_{ult}/FS$	q_{ult} = Ultimate bearing capacity in kPa, N = SPT Number, FS = Factor of Safety
Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition	
Deep Foundation	
For Cohesive Soil	
$f_{su} = \alpha C_u$; $q_{pu} = N_c C_u$	α = Adhesion factor, f_{su} = Ultimate skin friction, C_u = Undrained Shear Strength, q_{pu} = Ultimate end bearing of pile, N_c = Bearing capacity factor for deep foundation = 9
For Cohesion less Soil	
$f_{su} = N/50\ tsf$; $q_{pu} = 4N\ tsf$	f_{su} = Ultimate skin friction, q_{pu} = Ultimate end bearing of pile, N = SPT Number
Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"	

Based on the above formula, the important parameters for foundation design have been determined and are given in **Tables 7-11**.

Table 7A: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5) **

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square (Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH01	2.5-5.0	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	3	0.304	0.320	0.349	0.359	0.388	0.422	0.328
BH02	2.5-5.0	3	0.371	0.390	0.426	0.438	0.474	0.515	0.400
	5.0-10	3	0.312	0.328	0.358	0.368	0.398	0.433	0.336
BH03	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	4	0.512	0.538	0.587	0.605	0.653	0.711	0.552
BH04	2.5-5.0	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	3	0.388	0.408	0.445	0.458	0.495	0.539	0.418
BH05	2.5-5.0	3	0.302	0.317	0.346	0.356	0.385	0.419	0.325
	5.0-10	3	0.388	0.408	0.445	0.458	0.495	0.539	0.418
BH06	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	4	0.472	0.496	0.541	0.557	0.602	0.655	0.509
BH07	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH08	2.5-5.0	3	0.394	0.415	0.452	0.466	0.503	0.547	0.425
	5.0-10	20	2.268	2.385	2.600	2.679	2.894	3.148	2.444
BH09	2.5-5.0	3	0.394	0.415	0.452	0.466	0.503	0.547	0.425
	5.0-10	3	0.384	0.404	0.440	0.454	0.490	0.533	0.414
BH10	2.5-5.0	3	0.302	0.317	0.346	0.356	0.385	0.419	0.325
	5.0-10	3	0.392	0.412	0.449	0.463	0.500	0.544	0.422

Table 7B: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5) **

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH11	2.5-5.0	3	0.302	0.317	0.346	0.356	0.385	0.419	0.325
	5.0-10	4	0.472	0.496	0.541	0.557	0.602	0.655	0.509
BH12	2.5-5.0	4	0.418	0.439	0.479	0.493	0.533	0.580	0.450
	5.0-10	4	0.468	0.492	0.537	0.553	0.597	0.650	0.504
BH13	2.5-5.0	3	0.348	0.366	0.399	0.411	0.444	0.483	0.375
	5.0-10	3	0.388	0.408	0.445	0.458	0.495	0.539	0.418
BH14	2.5-5.0	2	0.255	0.268	0.293	0.301	0.326	0.354	0.275
	5.0-10	3	0.352	0.370	0.404	0.416	0.449	0.489	0.379
BH15	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
BH16	2.5-5.0	3	0.302	0.317	0.346	0.356	0.385	0.419	0.325
	5.0-10	4	0.428	0.450	0.491	0.505	0.546	0.594	0.461
BH17	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH18	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	8	0.872	0.917	1.000	1.030	1.113	1.210	0.940
BH19	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH20	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293

Table 7C: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5) **

Bore Hole No.	Depth	Design N	Clay						Sand
			Continuous			Square(Isolated)			
	(ft)	-	D/B=0.25	D/B=0.5	D/B=1	D/B=0.25	D/B=0.5	D/B=1	
BH21	2.5-5.0	2	0.278	0.293	0.319	0.329	0.355	0.386	0.300
	5.0-10	3	0.312	0.328	0.358	0.368	0.398	0.433	0.336
BH22	2.5-5.0	2	0.232	0.244	0.266	0.274	0.296	0.322	0.250
	5.0-10	2	0.272	0.286	0.312	0.321	0.347	0.378	0.293
BH23	2.5-5.0	5	0.603	0.634	0.692	0.712	0.770	0.837	0.650
	5.0-10	9	1.072	1.127	1.229	1.266	1.368	1.488	1.155
BH24	2.5-5.0	3	0.394	0.415	0.452	0.466	0.503	0.547	0.425
	5.0-10	5	0.544	0.572	0.624	0.642	0.694	0.755	0.586
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	2.5-5.0	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	5.0-10	0	0.000	0.000	0.000	0.000	0.000	0.000	0.000

**** Design engineer should consider the settlement issue along with the tabulated bearing capacity for designing a shallow foundation.**

Table 8A: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.074	0.138	0.074	0.118	0.074	0.096	0.074	0.074	0.074	0.074
10	0.096	0.049	0.049	0.074	0.074	0.049	0.049	0.074	0.096	0.049
15	0.049	0.074	0.049	0.049	0.049	0.049	0.049	0.096	0.074	0.074
20	0.049	0.074	0.196	0.118	0.118	0.174	0.074	0.669	0.074	0.118
25	0.156	0.118	0.118	0.138	0.096	0.096	0.118	0.240	0.096	0.074
30	0.196	0.096	0.118	0.214	0.138	0.174	0.261	0.156	0.138	0.246
35	0.074	0.074	0.218	0.049	0.049	0.261	0.321	0.214	0.096	0.218
40	0.239	0.174	0.268	0.118	0.074	0.241	0.669	0.241	0.049	0.321
45	0.096	0.118	0.321	0.214	0.156	0.401	0.669	0.669	0.294	0.441
50	0.239	0.218	0.375	0.334	0.281	0.415	0.669	0.669	0.495	0.230
55	0.261	0.268	0.682	0.375	0.468	0.669	0.669	0.669	0.696	0.268
60	0.669	0.682	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.401
65	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669
70	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 8B: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH11	BH12	BH13	BH14	BH15	BH16	BH17	BH18	BH19	BH20
5	0.074	0.096	0.074	0.074	0.049	0.074	0.049	0.049	0.049	0.049
10	0.049	0.074	0.074	0.049	0.049	0.074	0.049	0.049	0.049	0.049
15	0.074	0.096	0.074	0.049	0.049	0.049	0.049	0.049	0.074	0.049
20	0.156	0.118	0.096	0.118	0.049	0.138	0.074	0.144	0.049	0.074
25	0.096	0.096	0.230	0.074	0.196	0.096	0.074	0.156	0.196	0.074
30	0.049	0.074	0.226	0.074	0.184	0.118	0.240	0.144	0.196	0.218
35	0.118	0.096	0.214	0.214	0.168	0.156	0.239	0.144	0.241	0.294
40	0.096	0.136	0.230	0.240	0.160	0.168	0.128	0.160	0.254	0.230
45	0.152	0.136	0.184	0.294	0.208	0.176	0.176	0.160	0.375	0.128
50	0.160	0.160	0.200	0.144	0.200	0.176	0.200	0.176	0.200	0.144
55	0.192	0.128	0.192	0.160	0.192	0.168	0.192	0.192	0.192	0.208
60	0.184	0.144	0.184	0.184	0.184	0.184	0.184	0.184	0.184	0.184
65	0.184	0.184	0.184	0.184	0.184	-	0.184	0.184	0.184	0.184
70	0.176	0.176	0.176	0.176	0.176	-	0.176	0.176	0.176	0.176
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 8C: The allowable skin friction for Precast driven piles in Tsf (FS=2.5)

Depth	Bore Hole									
ft	BH21	BH22	BH23	BH24						
5	0.049	0.049	0.049	0.074	-	-	-	-	-	-
10	0.049	0.049	0.049	0.096	-	-	-	-	-	-
15	0.074	0.049	0.218	0.074	-	-	-	-	-	-
20	0.074	0.074	0.246	0.156	-	-	-	-	-	-
25	0.096	0.032	0.096	0.218	-	-	-	-	-	-
30	0.074	0.048	0.196	0.334	-	-	-	-	-	-
35	0.136	0.136	0.156	0.218	-	-	-	-	-	-
40	0.160	0.168	0.226	0.321	-	-	-	-	-	-
45	0.120	0.176	0.468	0.468	-	-	-	-	-	-
50	0.120	0.176	0.415	0.455	-	-	-	-	-	-
55	0.136	0.184	0.468	0.168	-	-	-	-	-	-
60	0.152	0.184	0.669	0.184	-	-	-	-	-	-
65	0.184	0.184	0.669	0.184	-	-	-	-	-	-
70	0.176	0.176	0.669	0.176	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9A: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.675	1.350	0.675	1.125	0.675	0.900	0.675	0.675	0.675	0.675
10	0.900	0.450	0.450	0.675	0.675	0.450	0.450	0.675	0.900	0.450
15	0.450	0.675	0.450	0.450	0.450	0.450	0.450	0.900	0.675	0.675
20	0.450	0.675	2.025	1.125	1.125	1.800	0.675	11.250	0.675	1.125
25	1.575	1.125	1.125	1.350	0.900	0.900	1.125	3.825	0.900	0.675
30	2.025	0.900	1.125	2.925	1.350	1.800	2.700	1.575	1.350	3.375
35	0.675	0.675	2.250	0.450	0.450	2.700	5.400	2.925	0.900	2.250
40	2.475	1.800	4.500	1.125	0.675	4.050	11.250	4.050	0.450	5.400
45	0.900	1.125	5.400	2.925	1.575	6.750	11.250	11.250	4.950	7.425
50	2.475	2.250	6.300	5.625	4.725	6.975	11.250	11.250	8.325	3.150
55	2.700	4.500	11.475	6.300	7.875	11.250	11.250	11.250	11.700	4.500
60	11.250	11.475	11.250	11.250	11.250	11.250	11.250	11.250	11.250	6.750
65	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250
70	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250	11.250
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9B: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH11	BH12	BH13	BH14	BH15	BH16	BH17	BH18	BH19	BH20
5	0.675	0.900	0.675	0.675	0.450	0.675	0.450	0.450	0.450	0.450
10	0.450	0.675	0.675	0.450	0.450	0.675	0.450	0.450	0.450	0.450
15	0.675	0.900	0.675	0.450	0.450	0.450	0.450	0.450	0.675	0.450
20	1.575	1.125	0.900	1.125	0.450	1.350	0.675	28.800	0.450	0.675
25	0.900	0.900	3.150	0.675	2.025	0.900	0.675	1.575	2.025	0.675
30	0.450	0.675	3.600	0.675	36.800	1.125	3.825	28.800	2.025	2.250
35	1.125	0.900	2.925	2.925	33.600	1.575	2.475	28.800	4.050	4.950
40	0.900	27.200	3.150	3.825	32.000	33.600	25.600	32.000	4.275	3.150
45	30.400	27.200	36.800	4.950	41.600	35.200	35.200	32.000	6.300	25.600
50	32.000	32.000	40.000	28.800	40.000	35.200	40.000	35.200	40.000	28.800
55	38.400	25.600	38.400	32.000	38.400	33.600	38.400	38.400	38.400	41.600
60	36.800	28.800	36.800	36.800	36.800	36.800	36.800	36.800	36.800	36.800
65	36.800	36.800	36.800	36.800	36.800	-	36.800	36.800	36.800	36.800
70	35.200	35.200	35.200	35.200	35.200	-	35.200	35.200	35.200	35.200
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 9C: The allowable end bearing for Precast driven piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH21	BH22	BH23	BH24						
5	0.450	0.450	0.450	0.675	-	-	-	-	-	-
10	0.450	0.450	0.450	0.900	-	-	-	-	-	-
15	0.675	0.450	2.250	0.675	-	-	-	-	-	-
20	0.675	0.675	3.375	1.575	-	-	-	-	-	-
25	0.900	6.400	0.900	2.250	-	-	-	-	-	-
30	0.675	9.600	2.025	5.625	-	-	-	-	-	-
35	27.200	27.200	1.575	2.250	-	-	-	-	-	-
40	32.000	33.600	3.600	5.400	-	-	-	-	-	-
45	24.000	35.200	7.875	7.875	-	-	-	-	-	-
50	24.000	35.200	6.975	7.650	-	-	-	-	-	-
55	27.200	36.800	7.875	33.600	-	-	-	-	-	-
60	30.400	36.800	11.250	36.800	-	-	-	-	-	-
65	36.800	36.800	11.250	36.800	-	-	-	-	-	-
70	35.200	35.200	11.250	35.200	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10A: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.048	0.090	0.048	0.077	0.048	0.062	0.048	0.048	0.048	0.048
10	0.062	0.032	0.032	0.048	0.048	0.032	0.032	0.048	0.062	0.032
15	0.032	0.048	0.032	0.032	0.032	0.032	0.032	0.062	0.048	0.048
20	0.032	0.048	0.127	0.077	0.077	0.113	0.048	0.435	0.048	0.077
25	0.101	0.077	0.077	0.090	0.062	0.062	0.077	0.156	0.062	0.048
30	0.127	0.062	0.077	0.139	0.090	0.113	0.170	0.101	0.090	0.160
35	0.048	0.048	0.141	0.032	0.032	0.170	0.209	0.139	0.062	0.141
40	0.156	0.113	0.174	0.077	0.048	0.156	0.435	0.156	0.032	0.209
45	0.062	0.077	0.209	0.139	0.101	0.261	0.435	0.435	0.191	0.287
50	0.156	0.141	0.243	0.217	0.183	0.270	0.435	0.435	0.322	0.149
55	0.170	0.174	0.443	0.243	0.304	0.435	0.435	0.435	0.452	0.174
60	0.435	0.443	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.261
65	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435
70	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10B: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH11	BH12	BH13	BH14	BH15	BH16	BH17	BH18	BH19	BH20
5	0.048	0.062	0.048	0.048	0.032	0.048	0.032	0.032	0.032	0.032
10	0.032	0.048	0.048	0.032	0.032	0.048	0.032	0.032	0.032	0.032
15	0.048	0.062	0.048	0.032	0.032	0.032	0.032	0.032	0.048	0.032
20	0.101	0.077	0.062	0.077	0.032	0.090	0.048	0.094	0.032	0.048
25	0.062	0.062	0.149	0.048	0.127	0.062	0.048	0.101	0.127	0.048
30	0.032	0.048	0.147	0.048	0.120	0.077	0.156	0.094	0.127	0.141
35	0.077	0.062	0.139	0.139	0.109	0.101	0.156	0.094	0.156	0.191
40	0.062	0.088	0.149	0.156	0.104	0.109	0.083	0.104	0.165	0.149
45	0.099	0.088	0.120	0.191	0.135	0.114	0.114	0.104	0.243	0.083
50	0.104	0.104	0.130	0.094	0.130	0.114	0.130	0.114	0.130	0.094
55	0.125	0.083	0.125	0.104	0.125	0.109	0.125	0.125	0.125	0.135
60	0.120	0.094	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
65	0.120	0.120	0.120	0.120	0.120	-	0.120	0.120	0.120	0.120
70	0.114	0.114	0.114	0.114	0.114	-	0.114	0.114	0.114	0.114
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 10C: The allowable skin friction for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH21	BH22	BH23	BH24						
5	0.032	0.032	0.032	0.048	-	-	-	-	-	-
10	0.032	0.032	0.032	0.062	-	-	-	-	-	-
15	0.048	0.032	0.141	0.048	-	-	-	-	-	-
20	0.048	0.048	0.160	0.101	-	-	-	-	-	-
25	0.062	0.021	0.062	0.141	-	-	-	-	-	-
30	0.048	0.031	0.127	0.217	-	-	-	-	-	-
35	0.088	0.088	0.101	0.141	-	-	-	-	-	-
40	0.104	0.109	0.147	0.209	-	-	-	-	-	-
45	0.078	0.114	0.304	0.304	-	-	-	-	-	-
50	0.078	0.114	0.270	0.296	-	-	-	-	-	-
55	0.088	0.120	0.304	0.109	-	-	-	-	-	-
60	0.099	0.120	0.435	0.120	-	-	-	-	-	-
65	0.120	0.120	0.435	0.120	-	-	-	-	-	-
70	0.114	0.114	0.435	0.114	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11A: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH01	BH02	BH03	BH04	BH05	BH06	BH07	BH08	BH09	BH10
5	0.304	0.608	0.304	0.506	0.304	0.405	0.304	0.304	0.304	0.304
10	0.405	0.203	0.203	0.304	0.304	0.203	0.203	0.304	0.405	0.203
15	0.203	0.304	0.203	0.203	0.203	0.203	0.203	0.405	0.304	0.304
20	0.203	0.304	0.911	0.506	0.506	0.810	0.304	5.063	0.304	0.506
25	0.709	0.506	0.506	0.608	0.405	0.405	0.506	1.721	0.405	0.304
30	0.911	0.405	0.506	1.316	0.608	0.810	1.215	0.709	0.608	1.519
35	0.304	0.304	1.013	0.203	0.203	1.215	2.430	1.316	0.405	1.013
40	1.114	0.810	2.025	0.506	0.304	1.823	5.063	1.823	0.203	2.430
45	0.405	0.506	2.430	1.316	0.709	3.038	5.063	5.063	2.228	3.341
50	1.114	1.013	2.835	2.531	2.126	3.139	5.063	5.063	3.746	1.418
55	1.215	2.025	5.164	2.835	3.544	5.063	5.063	5.063	5.265	2.025
60	5.063	5.164	5.063	5.063	5.063	5.063	5.063	5.063	5.063	3.038
65	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063
70	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063	5.063
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11B: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH11	BH12	BH13	BH14	BH15	BH16	BH17	BH18	BH19	BH20
5	0.304	0.405	0.304	0.304	0.203	0.304	0.203	0.203	0.203	0.203
10	0.203	0.304	0.304	0.203	0.203	0.304	0.203	0.203	0.203	0.203
15	0.304	0.405	0.304	0.203	0.203	0.203	0.203	0.203	0.304	0.203
20	0.709	0.506	0.405	0.506	0.203	0.608	0.304	12.960	0.203	0.304
25	0.405	0.405	1.418	0.304	0.911	0.405	0.304	0.709	0.911	0.304
30	0.203	0.304	1.620	0.304	16.560	0.506	1.721	12.960	0.911	1.013
35	0.506	0.405	1.316	1.316	15.120	0.709	1.114	12.960	1.823	2.228
40	0.405	12.240	1.418	1.721	14.400	15.120	11.520	14.400	1.924	1.418
45	13.680	12.240	16.560	2.228	18.720	15.840	15.840	14.400	2.835	11.520
50	14.400	14.400	18.000	12.960	18.000	15.840	18.000	15.840	18.000	12.960
55	17.280	11.520	17.280	14.400	17.280	15.120	17.280	17.280	17.280	18.720
60	16.560	12.960	16.560	16.560	16.560	16.560	16.560	16.560	16.560	16.560
65	16.560	16.560	16.560	16.560	16.560	-	16.560	16.560	16.560	16.560
70	15.840	15.840	15.840	15.840	15.840	-	15.840	15.840	15.840	15.840
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

Table 11C: The allowable end bearing for Bored piles in Tsf (FS = 2.5)

Depth	Bore Hole									
ft	BH21	BH22	BH23	BH24						
5	0.203	0.203	0.203	0.304	-	-	-	-	-	-
10	0.203	0.203	0.203	0.405	-	-	-	-	-	-
15	0.304	0.203	1.013	0.304	-	-	-	-	-	-
20	0.304	0.304	1.519	0.709	-	-	-	-	-	-
25	0.405	2.880	0.405	1.013	-	-	-	-	-	-
30	0.304	4.320	0.911	2.531	-	-	-	-	-	-
35	12.240	12.240	0.709	1.013	-	-	-	-	-	-
40	14.400	15.120	1.620	2.430	-	-	-	-	-	-
45	10.800	15.840	3.544	3.544	-	-	-	-	-	-
50	10.800	15.840	3.139	3.443	-	-	-	-	-	-
55	12.240	16.560	3.544	15.120	-	-	-	-	-	-
60	13.680	16.560	5.063	16.560	-	-	-	-	-	-
65	16.560	16.560	5.063	16.560	-	-	-	-	-	-
70	15.840	15.840	5.063	15.840	-	-	-	-	-	-
75	-	-	-	-	-	-	-	-	-	-
80	-	-	-	-	-	-	-	-	-	-
85	-	-	-	-	-	-	-	-	-	-
90	-	-	-	-	-	-	-	-	-	-
95	-	-	-	-	-	-	-	-	-	-
100	-	-	-	-	-	-	-	-	-	-

CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the proposed 14 storied residential cum commercial building.

Soil profile scenario

Overall soil information of the investigated site is very irregular in between the boreholes and vertical inconsistency exists in the soil formation. Different scenarios have been observed among the boreholes. Grey soft clay layer has been observed for topsoil followed by alternative layers has existed.

From the above discussion, it is very clear that the whole scenario of the proposed site is quite variable in nature.

Foundation options

In this situation, the foundation should be selected considering both the dimension and design load for the proposed building. A deep foundation (Bored/precast pile) may be selected to ensure the stability of the structure and to overcome the ground improvement complication. For designing any suitable deep foundation, the necessary data for skin friction and end bearing can be found in the attached table (8 to 11) in the report which is based on existing soil conditions.

Safety provisions

From the practical viewpoint, it has been found in many buildings that the structure failed due to soil settlement or judging the soil parameters incorrectly, providing no signs of structural damage. The earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of the foundation.

Attachments

The layout of the boreholes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix.

Special notes

These suggestions are absolutely tentative and the selection of type, depth, and dimensions of the foundation would be decided by the Geotechnical Consultant or Design Engineer. The client also performed direct shear test for boreholes no 5,7,12,22 and 24. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **(60-70)ft** as the client provided.

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